

Phase 2: Boundaries & Energy

When Fields Collide with Reality

Master Study Guide & Step-by-Step Derivations

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1 Introduction: The Nature of Materials

Up until now, our equations have relied heavily on ϵ_0 , the permittivity of free space (a perfect vacuum). But fields are often deployed inside physical materials.

In electromagnetic field theory, we broadly classify materials into two primary categories based on how their internal electrons behave when exposed to an external electric field:

- **Dielectrics (Insulators):** The electrons are tightly bound to their parent atoms. They cannot flow freely. However, an external electric field can "stretch" or "polarize" these atoms, slightly shifting the positive nucleus and the negative electron cloud. This internal polarization creates a weak secondary field that directly opposes the external field, effectively weakening it. We account for this by multiplying ϵ_0 by a relative permittivity factor, ϵ_r (also called the dielectric constant). The total permittivity is $\epsilon = \epsilon_0 \epsilon_r$.
- **Conductors (Metals):** The outermost valence electrons are entirely detached from their parent atoms and form a "sea" of free electrons. If an external electric field is applied, these free electrons experience a force ($\mathbf{F} = q\mathbf{E}$) and instantly rush to the surface of the material. They pile up on the surface until they generate their own internal field that *perfectly cancels out* the external field.

The Golden Rule of Perfect Conductors:

Because the free electrons move instantly to cancel any applied field, the net electric field strictly **inside** a perfect conductor in static equilibrium is always exactly zero ($\mathbf{E} = \mathbf{0}$). Consequently, the entire volume of a conductor acts as a single, massive **Equipotential Surface** (voltage is constant everywhere inside and on the surface).

2 Dielectric-Dielectric Boundary Conditions

The Problem: Imagine a beam of light (which is an electromagnetic wave) traveling through the air and hitting the surface of a glass pond. It bends (refracts). To mathematically predict how much it bends, we must know exactly how the electric field behaves at the exact microscopic boundary where the air (ϵ_1) touches the glass (ϵ_2).

We divide the Electric Field ($\mathbf{E}$) and the Electric Flux Density ($\mathbf{D}$) into two orthogonal components relative to the boundary plane:

1. **Tangential Component (E_t):** The part of the field swiping parallel along the surface.
2. **Normal Component (D_n):** The part of the field stabbing perpendicularly straight through the surface.

Let us derive the physical rules governing both.

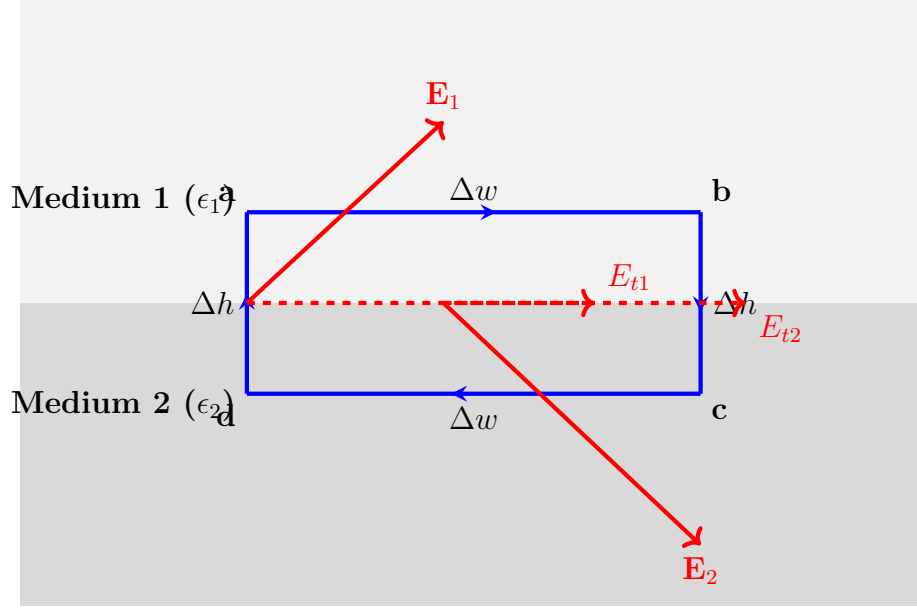
2.1 The Tangential Component (E_t)

The Concept (Building from knowns): In Phase 0 and Phase 1, we learned that the static electric field is conservative. This means that if you move a positive test charge around a closed loop and return to your starting point, the net electrical work done is exactly zero. Mathematically, this is expressed by the integral form of Faraday's Law for statics:

$$\oint_L \mathbf{E} \cdot d\mathbf{l} = 0 \quad (1)$$

We will simply draw a microscopic rectangular loop that is half inside Medium 1 and half inside Medium 2, and apply this exact law.

TikZ Diagram: Amperian/Faraday Loop at a Boundary



Step-by-Step Derivation:

Step 1: Set up the integral.

We break the total closed loop integral into four distinct line segments:

$$\oint_L \mathbf{E} \cdot d\mathbf{l} = \int_a^b \mathbf{E}_1 \cdot d\mathbf{l} + \int_b^c \mathbf{E} \cdot d\mathbf{l} + \int_c^d \mathbf{E}_2 \cdot d\mathbf{l} + \int_d^a \mathbf{E} \cdot d\mathbf{l} = 0 \quad (2)$$

Step 2: Apply the mathematical limit.

We want to know what happens *exactly* at the boundary. To do this, we take the mathematical limit as the height of our rectangular loop shrinks to zero ($\Delta h \rightarrow 0$), while keeping the width (Δw) finite and non-zero.

As the height collapses, the path length of the vertical segments ($b \rightarrow c$ and $d \rightarrow a$) becomes zero. Therefore, the work done along these vertical paths identically vanishes:

$$\lim_{\Delta h \rightarrow 0} \int_b^c \mathbf{E} \cdot d\mathbf{l} = 0 \quad \text{and} \quad \lim_{\Delta h \rightarrow 0} \int_d^a \mathbf{E} \cdot d\mathbf{l} = 0 \quad (3)$$

Step 3: Evaluate the horizontal segments.

For the top path ($a \rightarrow b$), the displacement vector $d\mathbf{l}$ is perfectly horizontal and has a length of Δw . The dot product $\mathbf{E}_1 \cdot d\mathbf{l}$ isolates exactly the tangential component of the field, E_{t1} .

$$\int_a^b \mathbf{E}_1 \cdot d\mathbf{l} = E_{t1} \Delta w \quad (4)$$

For the bottom path ($c \rightarrow d$), the displacement vector is moving backward (from right

to left). Therefore, the displacement is $-\Delta w$.

$$\int_c^d \mathbf{E}_2 \cdot d\mathbf{l} = -E_{t2}\Delta w \quad (5)$$

Step 4: Finalize the equation.

Substitute these evaluated horizontal paths back into the original closed loop equation:

$$(E_{t1}\Delta w) + 0 + (-E_{t2}\Delta w) + 0 = 0 \quad (6)$$

$$(E_{t1} - E_{t2})\Delta w = 0 \quad (7)$$

Because we stated that the width of the loop Δw is finite and non-zero, the only way for this mathematical equation to evaluate to zero is if the term inside the parenthesis is exactly zero:

$$E_{t1} - E_{t2} = 0 \quad (8)$$

$$E_{t1} = E_{t2} \quad (9)$$

Physical Conclusion:

The tangential component of the electric field intensity ($\mathbf{E}$) is perfectly continuous across a dielectric boundary.

What about the Flux Density ($\mathbf{D}$)?

We know the constitutive relation is $\mathbf{D} = \epsilon\mathbf{E}$, which means $\mathbf{E} = \frac{\mathbf{D}}{\epsilon}$. If we substitute this into our derived conclusion:

$$\frac{D_{t1}}{\epsilon_1} = \frac{D_{t2}}{\epsilon_2} \quad (10)$$

Because the permittivities (ϵ_1 and ϵ_2) are different, the tangential components of the electric flux density are **discontinuous**.

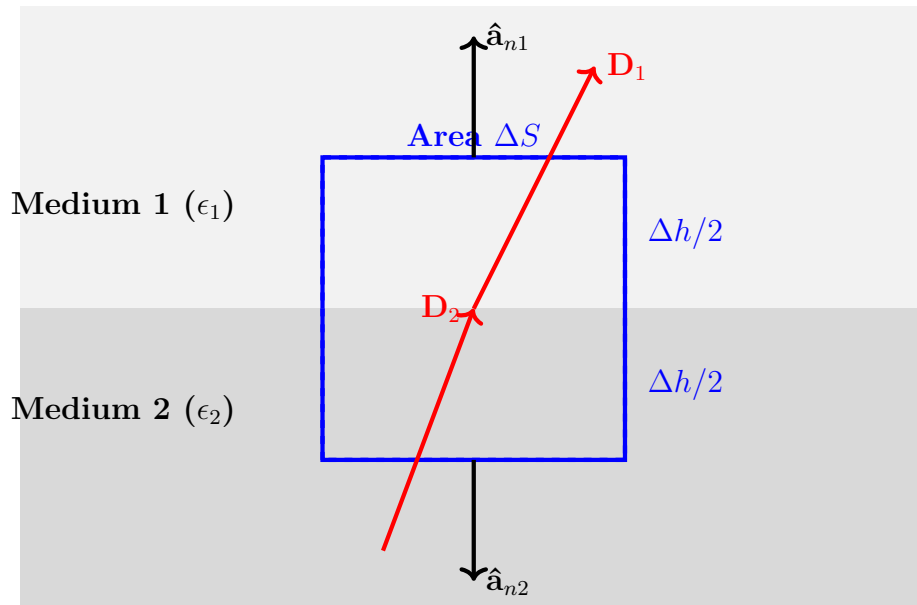
2.2 The Normal Component (D_n)

The Concept (Building from knowns): To find the normal (perpendicular) component, we rely on Gauss's Law. In Phase 1, we learned that the total electric flux leaving a closed surface equals the enclosed charge.

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = Q_{\text{enc}} \quad (11)$$

We will draw a microscopic cylindrical "pillbox" that punches directly through the boundary, and apply Gauss's Law.

TikZ Diagram: Gaussian Pillbox at a Boundary



Step-by-Step Derivation:

Step 1: Set up the integral.

We break the total closed surface integral into three distinct parts: the top flat circular face, the bottom flat circular face, and the curved cylindrical side wall.

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_{\text{top}} \mathbf{D}_1 \cdot d\mathbf{S} + \int_{\text{bottom}} \mathbf{D}_2 \cdot d\mathbf{S} + \int_{\text{sides}} \mathbf{D} \cdot d\mathbf{S} = Q_{\text{enc}} \quad (12)$$

Step 2: Apply the mathematical limit.

Just like before, we want to evaluate the field *exactly* at the boundary plane. We take the limit as the height of the pillbox shrinks to zero ($\Delta h \rightarrow 0$), while keeping the circular cross-sectional area (ΔS) finite.

As the height collapses, the surface area of the curved side walls becomes strictly zero. Therefore, no flux can physically exit through the sides:

$$\lim_{\Delta h \rightarrow 0} \int_{\text{sides}} \mathbf{D} \cdot d\mathbf{S} = 0 \quad (13)$$

Step 3: Evaluate the top and bottom faces.

The area vector $d\mathbf{S}$ always points outward, away from the enclosed volume. For the top face (in Medium 1), the outward normal is $\hat{\mathbf{a}}_{n1}$ (pointing UP). For the bottom face (in Medium 2), the outward normal is $\hat{\mathbf{a}}_{n2}$ (pointing DOWN).

Let us define a single, global reference normal vector $\hat{\mathbf{a}}_n$ that points upwards from Medium 2 into Medium 1. Consequently, $\hat{\mathbf{a}}_{n1} = \hat{\mathbf{a}}_n$, and $\hat{\mathbf{a}}_{n2} = -\hat{\mathbf{a}}_n$.

We assume the area ΔS is small enough that the flux density is uniform across it. The integrals simplify to simple dot products:

$$\int_{\text{top}} \mathbf{D}_1 \cdot d\mathbf{S} = \mathbf{D}_1 \cdot (\hat{\mathbf{a}}_{n1} \Delta S) = \mathbf{D}_1 \cdot (\hat{\mathbf{a}}_n \Delta S) \quad (14)$$

$$\int_{\text{bottom}} \mathbf{D}_2 \cdot d\mathbf{S} = \mathbf{D}_2 \cdot (\hat{\mathbf{a}}_{n2} \Delta S) = \mathbf{D}_2 \cdot (-\hat{\mathbf{a}}_n \Delta S) \quad (15)$$

Step 4: Define the enclosed charge.

As the pillbox flattens, the only volume left is the two-dimensional boundary plane itself. If there is any free charge trapped exactly on this boundary surface, it is defined by a surface charge density ρ_s (in Coulombs/m²). The total charge trapped inside the squashed pillbox is:

$$Q_{\text{enc}} = \rho_s \Delta S \quad (16)$$

Step 5: Finalize the equation.

Substitute all these evaluated parts back into Gauss's Law:

$$(\mathbf{D}_1 \cdot \hat{\mathbf{a}}_n) \Delta S - (\mathbf{D}_2 \cdot \hat{\mathbf{a}}_n) \Delta S = \rho_s \Delta S \quad (17)$$

$$(D_{n1} - D_{n2}) \Delta S = \rho_s \Delta S \quad (18)$$

Because the surface area ΔS is a finite, non-zero constant, we can divide it out of the equation perfectly:

$$D_{n1} - D_{n2} = \rho_s \quad (19)$$

Physical Conclusion:

For a standard ideal dielectric-dielectric interface, there are generally no free electrons available to form a sheet of surface charge ($\rho_s = 0$). In this vast majority of cases, the equation reduces to:

$$D_{n1} = D_{n2} \quad (20)$$

The normal component of the electric flux density ($\mathbf{D}$) is perfectly continuous across a charge-free dielectric boundary.

What about the Electric Field ($\mathbf{E}$)?

Using the constitutive relation $\mathbf{D} = \epsilon \mathbf{E}$:

$$\epsilon_1 E_{n1} = \epsilon_2 E_{n2} \tag{21}$$

Because the permittivities differ, the normal component of the electric field intensity is abruptly **discontinuous**.

3 Conductor-Dielectric Boundaries

Now, let us examine what happens when an electric field traveling through a dielectric (like air) hits the surface of a perfect metallic conductor (like a copper plate).

3.1 Applying the Boundary Rules

Recall our "Golden Rule" from the Introduction: The electric field strictly *inside* a perfect conductor is identically zero ($\mathbf{E}_2 = 0$, $\mathbf{D}_2 = 0$).

We can take the boundary conditions we just meticulously derived for two dielectrics and apply them here, simply by setting all Medium 2 parameters to zero.

1. The Tangential Condition:

$$E_{t1} = E_{t2} \quad (22)$$

$$E_{t1} = 0 \quad (23)$$

Conclusion: An electric field cannot swipe tangentially along the surface of a perfect conductor. If it did, it would instantly push the free electrons along the surface until they piled up and created an opposing field to kill that tangential push. Therefore, the electric field *must* hit a conductor surface perfectly perpendicularly (normally).

2. The Normal Condition:

$$D_{n1} - D_{n2} = \rho_s \quad (24)$$

$$D_{n1} - 0 = \rho_s \quad (25)$$

$$D_{n1} = \rho_s \quad \implies \quad \epsilon_1 E_{n1} = \rho_s \quad (26)$$

Conclusion: The normal electric field just outside the surface is directly and entirely caused by the surface charge density ρ_s residing on that exact metallic surface.

3.2 Highly Tested PYQ: Field "On" vs "Just Off" a Conductor

The Concept: A classic trick question in university exams (appearing in 2022 and 2024) asks you to justify the following statement: *"The electric field intensity exactly ON the conductor surface is half of the electric field intensity just OFF the conductor surface."*

Step-by-Step Logical Proof:

Step 1: The Macroscopic View (Just OFF the surface)

We just derived via the Gaussian pillbox that the total, macroscopic electric field hovering

microscopically just outside the conductor surface is:

$$E_{\text{off}} = \frac{\rho_s}{\epsilon} \quad (27)$$

(Note: Often, surface charge density is denoted by σ instead of ρ_s . We will use σ for the remainder of this proof to match standard textbook notation: $E_{\text{off}} = \frac{\sigma}{\epsilon}$).

Step 2: The Microscopic View (Exactly ON the surface)

Now, shrink down to the size of an atom and stand exactly on the physical surface of the conductor. We virtually partition the total charge of the entire conductor into two distinct groups:

1. A tiny, infinitesimally small circular "patch" of charge exactly where we are standing.
2. The rest of the charge distributed over the entire remainder of the massive conductor.

By the Principle of Superposition, the total field E_{off} must be the simple algebraic sum of the field created by the tiny patch (E_{patch}) and the field created by the rest of the conductor (E_{rest}).

$$E_{\text{off}} = E_{\text{patch}} + E_{\text{rest}} \quad (28)$$

If we are standing infinitesimally close to the tiny patch of charge, it mathematically behaves identically to an infinite flat sheet of charge. From introductory physics, the electric field generated by an infinite flat sheet radiates symmetrically outward from both sides with a magnitude of:

$$E_{\text{patch}} = \frac{\sigma}{2\epsilon} \quad (29)$$

If we know the total field is $\frac{\sigma}{\epsilon}$, and the patch contributes $\frac{\sigma}{2\epsilon}$, simple algebra reveals what the rest of the conductor must be contributing:

$$\frac{\sigma}{\epsilon} = \frac{\sigma}{2\epsilon} + E_{\text{rest}} \quad (30)$$

$$E_{\text{rest}} = \frac{\sigma}{\epsilon} - \frac{\sigma}{2\epsilon} = \frac{\sigma}{2\epsilon} \quad (31)$$

Step 3: The Crucial Physical Insight

Here is the core logical leap. We want to know the electric field experienced by the physical charge residing *exactly on* that tiny patch.

By Newton's Third Law, a physical object cannot exert a net force upon itself. You cannot lift yourself up by pulling your own shoelaces. Therefore, the charge in the tiny patch **does not feel its own electric field**.

The only electric field capable of exerting an actual, physical force on the charges

located within that specific patch is the field radiating from the *rest* of the conductor (E_{rest}).

Therefore, the active, driving electric field experienced exactly *on* the surface layer is:

$$E_{\text{on}} = E_{\text{rest}} = \frac{\sigma}{2\epsilon} \quad (32)$$

Comparing this to the total field just outside:

$$E_{\text{on}} = \frac{1}{2} \left(\frac{\sigma}{\epsilon} \right) = \frac{1}{2} E_{\text{off}} \quad (33)$$

This mathematically and physically justifies the exam statement.

3.3 Mechanical Pressure on a Conductor Surface

Because the surface charges are physically experiencing this electric field (E_{on}), they experience a mechanical force pushing them outward (since like charges repel each other, they try to escape the conductor).

Let us calculate this outward electrostatic force dF on our tiny differential patch of area dS . The total charge on our tiny patch is $dq = \sigma dS$. The physical force is the charge multiplied by the field it experiences:

$$dF = dq \cdot E_{\text{on}} \quad (34)$$

$$dF = (\sigma dS) \cdot \left(\frac{\sigma}{2\epsilon} \right) \quad (35)$$

$$dF = \frac{\sigma^2}{2\epsilon} dS \quad (36)$$

Mechanical Pressure (P) is defined fundamentally as force per unit area ($P = dF/dS$). We divide the differential force by the differential area of the patch:

$$P = \frac{dF}{dS} = \frac{\sigma^2}{2\epsilon} \quad [\text{N/m}^2] \quad (37)$$

We can make this formula more elegant by substituting the macroscopic electric field $E_{\text{off}} = \frac{\sigma}{\epsilon}$, which means $\sigma = \epsilon E_{\text{off}}$. Let's just call it E for simplicity.

$$P = \frac{(\epsilon E)^2}{2\epsilon} \quad (38)$$

$$P = \frac{\epsilon^2 E^2}{2\epsilon} \quad (39)$$

$$P = \frac{1}{2} \epsilon E^2 \quad [\text{N/m}^2] \quad (40)$$

This profound equation shows that an electric field effectively "pulls" on the surface of a conductor, exerting a physical mechanical tension (pressure) that grows with the square of the field strength.

4 Electrostatic Energy Density

The Problem: We know that a capacitor stores energy. But *where* is that energy stored? Is it located on the metal plates? Or is it stored invisibly in the empty vacuum space between the plates? Field theory argues that energy resides fundamentally within the field itself.

Let us rigorously derive the expression for the energy density (Joules per cubic meter) stored dynamically in an electrostatic field.

4.1 Derivation 1: From Assembling Charges

(Note: This specific derivation path is heavily tested in the 2023 exam).

Step 1: Energy of a Discrete Charge Distribution

Consider an entirely empty, infinite universe. We bring in a first discrete point charge q_1 from absolute infinity to a specific spatial point P_1 . Since there is absolutely no existing electric field to fight against, it requires exactly zero mechanical work ($W_1 = 0$).

Next, we bring in a second discrete point charge q_2 from infinity to a point P_2 . This incoming charge must do mechanical work against the established potential $V_{1,2}$ created by the presence of q_1 .

$$W_2 = q_2 V_{1,2} \quad (41)$$

Bringing in a third point charge q_3 to point P_3 requires doing work against the combined potentials of both q_1 and q_2 :

$$W_3 = q_3 V_{1,3} + q_3 V_{2,3} \quad (42)$$

The total accumulated potential energy W_E stored in a macroscopic assembly of N point charges is the sum of all this required external work:

$$W_E = \frac{1}{2} \sum_{k=1}^N q_k V_k \quad (43)$$

where V_k is the total composite electric potential at the exact location of charge q_k generated by all other charges. The $\frac{1}{2}$ mathematical factor is critical; it corrects for double-counting the pairwise interaction energy (since the energetic interaction between charge 1 and 2 is identically the same as the interaction between charge 2 and 1).

Step 2: Transition to Continuous Charge Distributions

For a continuous macroscopic volume charge distribution (rather than discrete, sparse points), we mathematically replace the discrete point charge q_k with an infinitesimal differential charge element $dq = \rho_v dv$.

We then replace the discrete mathematical summation operator with a continuous volume integral calculated over all space V where the physical charge exists:

$$W_E = \frac{1}{2} \iiint_{\text{vol}} \rho_v V dv \quad (44)$$

where ρ_v is the continuous volume charge density, and V is the spatial scalar electric potential function.

Step 3: Implementing Maxwell's Equations and Vector Identities

We want to express this energy purely in terms of fields, not charges. We rigorously substitute the point differential form of Gauss's Law ($\rho_v = \nabla \cdot \mathbf{D}$) directly into Equation 44:

$$W_E = \frac{1}{2} \iiint_{\text{vol}} (\nabla \cdot \mathbf{D}) V dv \quad (45)$$

To manipulate this complex expression algebraically, we deploy a standard vector calculus product rule identity determining the spatial divergence of a scalar V multiplied by a vector field $\mathbf{D}$:

$$\nabla \cdot (\mathbf{V}\mathbf{D}) = V(\nabla \cdot \mathbf{D}) + \mathbf{D} \cdot (\nabla V) \quad (46)$$

Rearranging this identity to cleanly isolate the exact parenthetical term we need:

$$V(\nabla \cdot \mathbf{D}) = \nabla \cdot (\mathbf{V}\mathbf{D}) - \mathbf{D} \cdot (\nabla V) \quad (47)$$

We substitute this rearranged vector identity back into our primary energy volume integral:

$$W_E = \frac{1}{2} \iiint_{\text{vol}} [\nabla \cdot (\mathbf{V}\mathbf{D}) - \mathbf{D} \cdot (\nabla V)] dv \quad (48)$$

By the linearity of integration, we split this into two entirely separate volume integrals:

$$W_E = \frac{1}{2} \iiint_{\text{vol}} \nabla \cdot (\mathbf{V}\mathbf{D}) dv - \frac{1}{2} \iiint_{\text{vol}} \mathbf{D} \cdot (\nabla V) dv \quad (49)$$

Step 4: Applying the Divergence Theorem and Boundary Limits

We apply the Divergence Theorem (Gauss's Theorem) exclusively to the first integral term of Equation 49 to convert the volume integral of a divergence into a closed surface integral bounding that exact volume:

$$\iiint_{\text{vol}} \nabla \cdot (\mathbf{V}\mathbf{D}) dv = \oint_S (\mathbf{V}\mathbf{D}) \cdot d\mathbf{S} \quad (50)$$

To account for absolutely all the energy stored in the universe, the integration volume must theoretically extend out to infinity (meaning we define a spherical boundary with radius $r \rightarrow \infty$).

For any localized, finite charge distribution, at extreme distances approaching infinity, the scalar potential V falls off mathematically proportional to $1/r$. The flux density vector field $\mathbf{D}$ falls off proportional to $1/r^2$. Therefore, their combined scalar product $(V\mathbf{D})$ falls off exponentially fast as $1/r^3$. However, the geometric surface area of the bounding sphere at infinity S only grows proportionally as r^2 .

Let us evaluate the limit at infinity:

$$\lim_{r \rightarrow \infty} \oint_S (V\mathbf{D}) \cdot d\mathbf{S} \propto \left(\frac{1}{r^3}\right) \times r^2 = \frac{1}{r} \rightarrow 0 \quad (51)$$

Thus, as the integration volume encompasses all of space, the first boundary term completely and mathematically vanishes to zero.

Step 5: Finalizing the Energy Density Expression

We are left solely with the second integral term of Equation 49:

$$W_E = -\frac{1}{2} \iiint_{\text{all space}} \mathbf{D} \cdot (\nabla V) dv \quad (52)$$

Recall the fundamental foundational relationship connecting the electric field intensity and the scalar potential: $\mathbf{E} = -\nabla V$. We substitute this crucial relation directly into the integral:

$$\begin{aligned} W_E &= -\frac{1}{2} \iiint_{\text{all space}} \mathbf{D} \cdot (-\mathbf{E}) dv \\ W_E &= \iiint_{\text{all space}} \left(\frac{1}{2} \mathbf{D} \cdot \mathbf{E}\right) dv \end{aligned} \quad (53)$$

The volume density of electrostatic energy, denoted as w_e , is defined mathematically as the raw integrand of the total energy volume integral (i.e., the localized energy per unit volume, dW_E/dv):

$$w_e = \frac{1}{2} \mathbf{D} \cdot \mathbf{E} \quad (54)$$

For a linear, isotropic, and homogeneous dielectric medium where the constitutive relation $\mathbf{D} = \epsilon\mathbf{E}$ holds true, this universally simplifies to:

$$w_e = \frac{1}{2}(\epsilon\mathbf{E}) \cdot \mathbf{E} = \frac{1}{2}\epsilon(\mathbf{E} \cdot \mathbf{E}) = \frac{1}{2}\epsilon|\mathbf{E}|^2 \quad (55)$$

4.2 Alternative Derivation: From Mechanical Pressure

(Note: This alternative derivation path is heavily tested in the 2022 exam).

We already proved in Equation 40 that the outward mechanical pressure (force per unit area) exerted by an electric field on the surface of a charged conductor is $P = \frac{1}{2}\epsilon E^2$.

Imagine a virtual thermodynamic process. Suppose the electrostatic force pushes the surface of the conductor outward by a tiny, differential distance dx . The mechanical work dW done by the field during this microscopic expansion is Force $\times$ Distance:

$$dW = F \cdot dx \quad (56)$$

$$dW = (P \cdot A) \cdot dx \quad (57)$$

$$dW = \left(\frac{1}{2}\epsilon E^2 \cdot A\right) \cdot dx \quad (58)$$

Notice the geometric term $(A \cdot dx)$. Because A is the surface area and dx is the perpendicular outward displacement, their product represents the differential volume dv that is newly created/swept out by the expanding surface:

$$dv = A \cdot dx \quad (59)$$

Substituting this into the work equation gives:

$$dW = \frac{1}{2}\epsilon E^2 dv \quad (60)$$

By the strict principle of conservation of energy, the mechanical work done by the field must come at the expense of the potential energy stored dynamically in the electrostatic field itself.

The energy stored per unit volume (Energy Density, w_e) is simply the work done divided by the volume created:

$$w_e = \frac{dW}{dv} = \frac{1}{2}\epsilon E^2 \quad [\text{Joules/m}^3] \quad (61)$$

Notice how both wildly different derivations—one tracking the microscopic assembly of individual charges from infinity, and one tracking the macroscopic thermodynamic expansion of a metal surface—converge on the exact same beautiful, elegant equation.

The universe is consistent.

5 Phase 2 Summary and Exam Checklist

Congratulations! You have successfully mastered the physics of how electric fields interact with physical materials. You have transitioned from ideal vacuums to the complex reality of boundaries and energy storage.

Before advancing to Phase 3 (The Heavy Lifters: Numerical Methods and Images), ensure you can confidently check off every item on this list:

- ☐ I can fully reproduce the line-integral derivation to prove that tangential electric fields are continuous across a boundary ($E_{t1} = E_{t2}$).
- ☐ I can fully reproduce the Gaussian pillbox derivation to prove that normal electric flux density is discontinuous by the surface charge ($D_{n1} - D_{n2} = \rho_s$).
- ☐ I understand the "Golden Rule" of perfect conductors: the internal electric field is zero, and the entire volume is an equipotential surface.
- ☐ I can explain the physical logic of why the electric field exactly *on* a conductor's surface is exactly half the magnitude of the field just *off* the surface.
- ☐ I can confidently derive the energy density formula $w_e = \frac{1}{2}\epsilon E^2$ using *both* the charge assembly method (using the Divergence Theorem) and the mechanical pressure method.

Once this checklist is complete, you are fully prepared to tackle the intense mathematical applications found in **Phase 3: FDM, Images, & Cables**.